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One new species and one new subspecies of the water mite genus *Hygrobates* Koch, 1837 from Cyprus (Acari, Hydrachnidia, Hygrobatidae)

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Abstract

Two new taxa of the water mite genus *Hygrobates* Koch, 1837 are described from Cyprus: *H. dorflingeri* n. sp. and *H. ponticus cyprioticus* n. ssp. Both belong to the *H. calliger* complex, whose members are common components of water mite assemblages in low-, middle-, and high-order streams. Phylogenetic analysis and ASAP species delimitation based on COI barcoding support their recognition as distinct lineages. Their morphological and genetic similarity to close relatives from mainland Anatolia suggests potential dispersal pathways for ancestral lineages during the colonization of Cyprus.

Key words new species, DNA barcoding, taxonomy, Mediterranean.

Introduction

Hygrobates Koch, 1837 is the second most species-rich genus in the family Hygrobatidae. In the Western Palearctic, the genus is represented by two subgenera: the polyacetabulate *Rivobates*, characterized by a genital field with more than three, often numerous, pairs of acetabula, and the nominate subgenus, which is distributed across all continents except Antarctica (Smit 2020).

Traditionally, in water mite literature, *Hygrobates* specimens with a coarsely lineated soft integument have been assigned to *H. calliger* Piersig, 1896. This species was considered a common inhabitant of low-, middle-, and high-order streams with a Palearctic distribution (Matsumoto *et al.* 2005; Gerecke *et al.* 2016). Pešić *et al.* (2022) first questioned the taxonomic status of *H. calliger* using an integrative approach based on standardized sequences of the cytochrome c oxidase subunit I (COI) gene. Their DNA barcoding study demonstrated that *H. calliger* represents a species complex consisting of several distinct, previously overlooked lineages. Some of these lineages (e.g., *H. calliger* Clades 1, 2,

and 5 *sensu* Pešić *et al.* 2022) inhabit large areas, whereas others (e.g., *H. calliger* Clade 3 from Calabria and *H. tyrrhenicus* Pešić & Smit, 2022 from Corsica) appear to have a more restricted distribution. Tracking the spatial patterns of these lineages using Barcode Index Numbers (BINs), which act as reliable molecular surrogates for true species, provides valuable insight into their geographic distribution and the potential presence of cryptic or pseudocryptic diversity.

In this study, we examined several samples belonging to the *Hygrobates calliger* complex collected in Cyprus. By combining morphological analysis with COI-based phylogenetic results, we describe and introduce herein one new species and one new subspecies.

Material and Methods

Water mites were collected during a field trip to Cyprus in May 2025. The specimens examined in this study were sampled from two sites within the Kouris River catchment on the southern slopes of the Troodos Massif. Detailed descriptions of these sampling localities are provided by Pešić *et al.* (2026c). Additional specimens were obtained by the third author during a macroinvertebrate sampling campaign aimed at assessing the ecological quality of Cypriot streams.

Specimens were collected using hand nets and immediately preserved in 96% ethanol for molecular analysis. Following non-destructive DNA extraction, specimen vouchers were stored in 96% ethanol for morphological examination. Selected vouchers were subsequently dissected and slide-mounted in Faure's medium, while the remaining specimens were transferred to Koenike's fluid for long-term storage. The holotypes and paratypes of the new species are deposited in the Naturalis Biodiversity Center, Leiden, The Netherlands (RMNH).

Morphological nomenclature follows Gerecke *et al.* (2016). All measurements are given in micrometers (μm). Genital acetabula are measured on both sides and their values are given as a range. The following abbreviations are used: Ac = acetabula; BS = bootstrap support; Cx-I = first coxae; dL = dorsal length; H = height; I-L-1-6 = first-sixth segment of first leg; L = length; mL = median length; n = number of specimens examined; P-1 to P-5 = palp segment 1-5; RMNH = Naturalis Biodiversity Center, Leiden; W = width.

DNA barcode analyses

Molecular analyses were conducted at the Canadian Centre for DNA Barcoding (CCDB; ccdb.ca) in Guelph, Canada. At this institution, the barcode region of the COI gene was sequenced using standard invertebrate DNA extraction, amplification, and sequencing protocols. In total, DNA was extracted from five specimens of the *Hygrobates calliger* complex (Table 1). Relevant voucher information and consensus sequences were deposited in the Barcode of Life Data Systems (BOLD) under the project DS-CYPHYD "DNA Barcodes of water mites of Cyprus" (dx.doi.org/10.5883/DS-CYPHYD).

Table 1. Details of DNA barcoded specimens, including localities and coordinates of sampling sites from Cyprus, and the barcode index number codes. BOLD data presented here was last accessed on 5th September 2026.

Locality	Coordinates	Sample ID	Process ID	BIN
<i>Hygrobates dorflingeri</i> n. sp.				
Cyprus, Limassol, Mesa	34.876° N,	CCDB_45908_A08	CYPTU008-25	BOLD:AH18373
potamos stream	32.910° E	CCDB_45908_B05	CYPTU017-25	
<i>Hygrobates ponticus cyprioticus</i> n. ssp.				
Cyprus, Limassol, Mesa	34.876° N,	CCDB_45908_B03	CYPTU015-25	BOLD:AH18370
potamos stream	32.910° E			
Stavros tis Psokas near	34.952576° N,	CCDB_45908_H02	CYPTU086-25	
Sarama Quarry	32.520718° E			
Agriokalami Coast (river mouth)	34.81257° N,	CCDB_45908_H06	CYPTU090-25	
	32.39443° E			

In addition to the sequences obtained in this study, additional COI sequence data of *Hygrobat* *calliger* complex were obtained from Pešić *et al.* (2022, 2023, 2026a) and downloaded from the BOLD Systems public database. Sequence alignments were performed using MUSCLE (Edgar 2004). Intra- and interspecific genetic distances were calculated based on the Kimura 2-parameter model (K2P; Kimura 1980) using MEGA 11 software (Tamura *et al.* 2021). The latter was also utilized to construct Maximum Likelihood (ML) trees. The best-fit substitution model (TN93+G+I) was selected using the Bayesian Information Criterion (BIC) implemented in MEGA 11. Tree construction incorporated an initial Neighbour-Joining (NJ) tree and utilized the Subtree-Pruning-Regrafting - Extensive heuristic search algorithm (SPR level 5). Branch support was calculated using the nonparametric bootstrap method (Felsenstein 1985) with 500 replicates, with values shown adjacent to the branches. Codon positions included were 1st+2nd+3rd+Noncoding.

Additionally, the sequence data were analyzed using the Assemble Species by Automatic Partitioning (ASAP), a method designed to species partitioning using a hierarchical clustering algorithm based on the pairwise distance distribution (Puillandre *et al.* 2021). We used the online ASAP version (<https://spartexplorer.mnhn.fr/>) with default settings and K2P distance model.

Results

Species delimitation using DNA-barcodes

The final alignment for species delimitation using COI sequence data included 57 sequences from the *Hygrobat* *calliger* complex, with *H. fluviatilis* (Ström, 1768) (NLACA1046-17) from the Netherlands used as the outgroup. This final alignment consisted of 1,221 nucleotide positions. The resulting ML tree is presented in Figure 1.

The COI phylogenetic analysis revealed nine highly supported, species-level clades within the studied *H. calliger* complex dataset. Six of these corresponded to the six previously known clades of the *H. calliger* complex identified by Pešić *et al.* (2022):

- *H. calliger* Clade 1 *sensu* Pešić *et al.* 2022 (BOLD:AEF4261, BOLD:AET1162, BOLD:AEO0848, BOLD:AFE0075), distributed from Portugal to Greece and northwards to Norway;
- *H. calliger* Clade 2 *sensu* Pešić *et al.* 2022 (BOLD:AEL5782), known from Germany and Russia southwards to Montenegro and Bulgaria;
- *H. calliger* Clade 3 *sensu* Pešić *et al.* 2022 (BOLD:AEK4718, BOLD:AEK4719, BOLD:AEK4720) from Calabria (Italy);
- *H. calliger* Clade 5 *sensu* Pešić *et al.* 2022 (BOLD:AEO5296), known from Germany and Russia southwards to Croatia;
- *Hygrobat* *tyrrhenicus* Pešić & Smit, 2022 (BOLD:AEO1422) from Corsica (France); and
- *H. ponticus* Pešić, Esen & Mumladze, 2022, known from Türkiye and Georgia.

Within the *H. ponticus* lineage, three subclades can be separated. The first subclade includes three specimens from Cyprus representing a unique BIN (BOLD:AH18370), described herein as *H. ponticus cyprioticus* **n. ssp.** This subclade is recovered as sister to the lineage (BOLD:AEO0649) that includes the nominotypical subspecies, *H. p. ponticus*, from eastern Türkiye and Georgia. The third subclade encompasses sequences of another subspecies, *H. ponticus yildirimi* Pešić, Zawal, Gölle, Cakmak & Saboori, 2023, which is known from two BINs: one from southwestern Türkiye (BOLD:AEY7516) and one from eastern Türkiye (BOLD:AGO8960). The seventh lineage, which includes two specimens from Cyprus described here as *H. dorflingeri* **n. sp.**, was revealed with a very strong support (BS = 99%) as the sister clade of *H. turani* Pešić, 2026 from eastern Türkiye. Finally, the ninth lineage encompasses sequences of *H. anaticus* Esen & Pešić, 2013 from eastern Türkiye (Esen *et al.* 2013), represented by two BINs (BOLD:AEX2126 and BOLD:AEY8569).

The average genetic distance between the clades recovered within the studied *Hygrobat* *calliger* complex dataset ranged from 12.28% (between *H. dorflingeri* **n. sp.** and *H. turani*) to 25.74% (between *H. dorflingeri* **n. sp.** and *H. calliger* Clade 1; Table 2). The average K2P distance between clades within *Hygrobat* *ponticus* varied from 8.64% (between *H. p. cyprioticus* **n. ssp.** and *H. p. ponticus*) to 10.25% (between *H. p. cyprioticus* **n. ssp.** and *H. p. yildirimi*). The ASAP delimitation method identified molecular operational taxonomic units (MOTUs) that mostly matched the nine

clusters recovered in the phylogenetic tree. The best ASAP score (3.50, at a threshold distance of 0.19% K2P distance) merged all three subclades of *H. ponticus* into a single MOTU. Conversely, the second-best ASAP score (4.50, at a threshold distance of 0.11% K2P distance) separated these three subclades into distinct MOTUs.

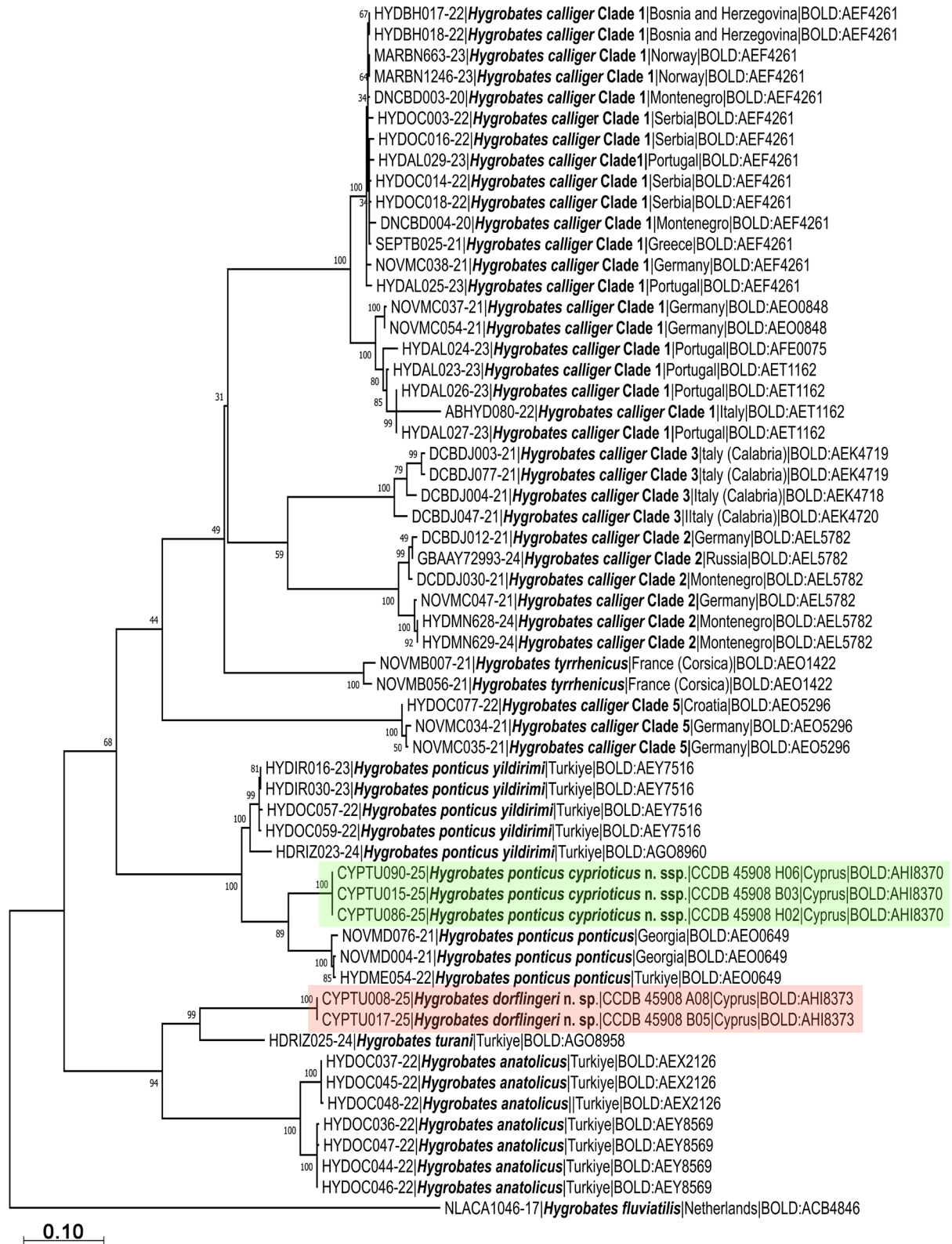


Figure 1. Maximum Likelihood tree (TN93+G+I model) of the *Hygrobatas calliger* complex obtained from 52 nucleotide COI sequences. Newly generated sequences are highlighted in colour: the *H. dorflingeri* n. sp. clade is marked in red, whereas *H. ponticus cyprioticus* n. ssp. is shown in green.

Table 2. Estimates of average genetic distance (K2P) within and between COI clades examined of the *Hygrobates calliger* complex.

Taxa	Intraspecific	Interspecific									
		(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
(1) <i>Hygrobates calliger</i> Clade 1	0.0379										
(2) <i>Hygrobates calliger</i> Clade 2	0.0226	0.2058									
(3) <i>Hygrobates calliger</i> Clade 3	0.0332	0.1936	0.1743								
(4) <i>Hygrobates calliger</i> Clade 5	0.0077	0.2223	0.2399	0.2330							
(5) <i>Hygrobates tyrrenicus</i>	0.0205	0.1987	0.1843	0.1984	0.2102						
(6) <i>Hygrobates ponticus ponticus</i>	0.0079	0.2178	0.2240	0.2472	0.2347	0.2183					
(7) <i>Hygrobates ponticus yildirimi</i>	0.0156	0.2104	0.1979	0.2240	0.2296	0.2109	0.1025				
(8) <i>Hygrobates ponticus cyprioticus</i> n. ssp.	0	0.2268	0.2045	0.2114	0.2258	0.2171	0.0864	0.1006			
(9) <i>Hygrobates turani</i>	n/c	0.2292	0.2412	0.2380	0.2314	0.2173	0.2090	0.2042	0.2166		
(10) <i>Hygrobates dorflingeri</i> n. sp.	0	0.2574	0.2406	0.2336	0.2541	0.2294	0.2213	0.2216	0.2270	0.1228	
(11) <i>Hygrobates anatolicus</i>	0.0239	0.2318	0.2251	0.2299	0.2709	0.2319	0.2194	0.2199	0.2408	0.1607	0.1771

Systematics

Family Hygrobatidae Koch, 1842

Genus *Hygrobates* Koch, 1837

Subgenus *Hygrobates* s.s.

Hygrobates dorflingeri Pešić n. sp.

<https://www.zoobank.org/urn:lsid:zoobank.org:act:3B9A0430-C509-491E-BAA2-3A67CCDB9B3C>

Figures 2-3

Material examined — Holotype ♂, dissected and slide mounted (RMNH), sequenced (voucher Id: CCDB_45908_B05, sequence Id: CYPTU017-25), Cyprus, Limassol, Mesa Potamos stream, 34.876° N, 32.910° E, permanent stream, 23 May 2025, leg. Pešić, Tziortzis & Papatheodoulou. Paratypes: 1 ♀ same data as the holotype, barcoded (voucher Id: CCDB_45908_A08, sequence Id: CYPTU008-25), dissected and slide mounted (RMNH).

Diagnosis — *Morphological*: Integument lineated; posteromedial margin of Cx-I in both sexes triangular and less projecting; anterior margin of male genital plate without a knob-shaped medial projection, male genital acetabula smaller in size, L Ac-2 and -3 < 100 µm. *Molecular*: This lineage is represented by a unique Barcode Index Number (BIN: **BOLD:AH18373**). It differs from its closest congeneric lineage, *Hygrobates turani*, by a 12.28% K2P genetic divergence.

Description. General features — Colour yellow to brown; integument soft, characteristic lineation occasionally visible (most of the integument appeared smooth, probably as a result of the molecular analysis treatment). Posteromedial margin of Cx-I obtuse-triangular (Figure 2A), Cx-IV medial margin nose-like protruding. Excretory pore unsclerotized. P-2 ventral margin slightly concave, distally forming a shortened projection, denticles covering two-thirds of the ventral margin of both P-2 and P-3; P-4 ventral setae on the same level. *Male* — Anterior margin of genital plate with nearly indistinct medial projection, posterior margin indented (Figure 2B); acetabula arranged in an obtuse triangle. *Female* — Genital plates with a narrow border of porose secondary sclerotization, gonopore longer than genital plates (Figure 3A).

Measurements. Male (holotype [CCDB_45908_B05]) — Idiosoma L 1120, W 925; coxal field: L 463; Cx-III 556; mL of Cx-I + gnathosoma L 392; distance between lateralmost ends of caudo-lateral Cx-II apodemes, 175. Genital field L/W 200/261, ratio 0.77, L Ac-1-3: 56-66, 78-81, 70. Ejaculatory complex L 222.

Palp — total L 536; dL/H: P-1, 39/50; P-2, 140/92; P-3, 114/80; P-4, 183/42; P-5, 60/20; L P-2/P-4 ratio 0.77.

Legs — dL of I-L-2-5: 106, 148, 215, 219. dL of IV-L-2-6: 163, 234, 331, 334, 291.

Female (paratype [CCDB_45908_A08]) — Idiosoma L 1215, W 980; coxal field: L 472; Cx-III 609; mL of Cx-I + gnathosoma L 392; distance between lateralmost ends of caudo-lateral Cx-II apodemes, 184; genital field L/W 258/325; genital plates L 188; pregenital sclerite W 100; gonopore L 209; L Ac 1-3: 71, 103, 84.

Palp — Total L 583; dL/H: P-1, 41/56; P-2, 150/95; P-3, 128/84; P-4, 198/44; P-5, 66/23; L P-2/P-4 ratio 0.76. Chelicera total L 400, basal segment L 278, claw L 143, L claw/basal segment ratio 0.51.

Legs — dL of I-L-1-6: 81, 119, 153, 219, 225, 203. dL of IV-L-1-6: 156, 178, 238, 338, 340, 291.

Etymology — The new species is named in honor of Dr. Gerald Dörflinger (Water Development Department, Cyprus) in appreciation of his extensive work on eco-hydrological monitoring, protection and management of the unique inland freshwaters of Cyprus.

DNA Barcode Analysis — The COI barcodes of the new species form a unique Barcode Index Number (BIN) **BOLD:AH18373**, which exhibits a 12.27% K2P divergence from its most morphologically and genetically similar congener, *Hygrobates turani*, known from the eastern Black Sea coast of Türkiye (see Pešić *et al.* 2026a).

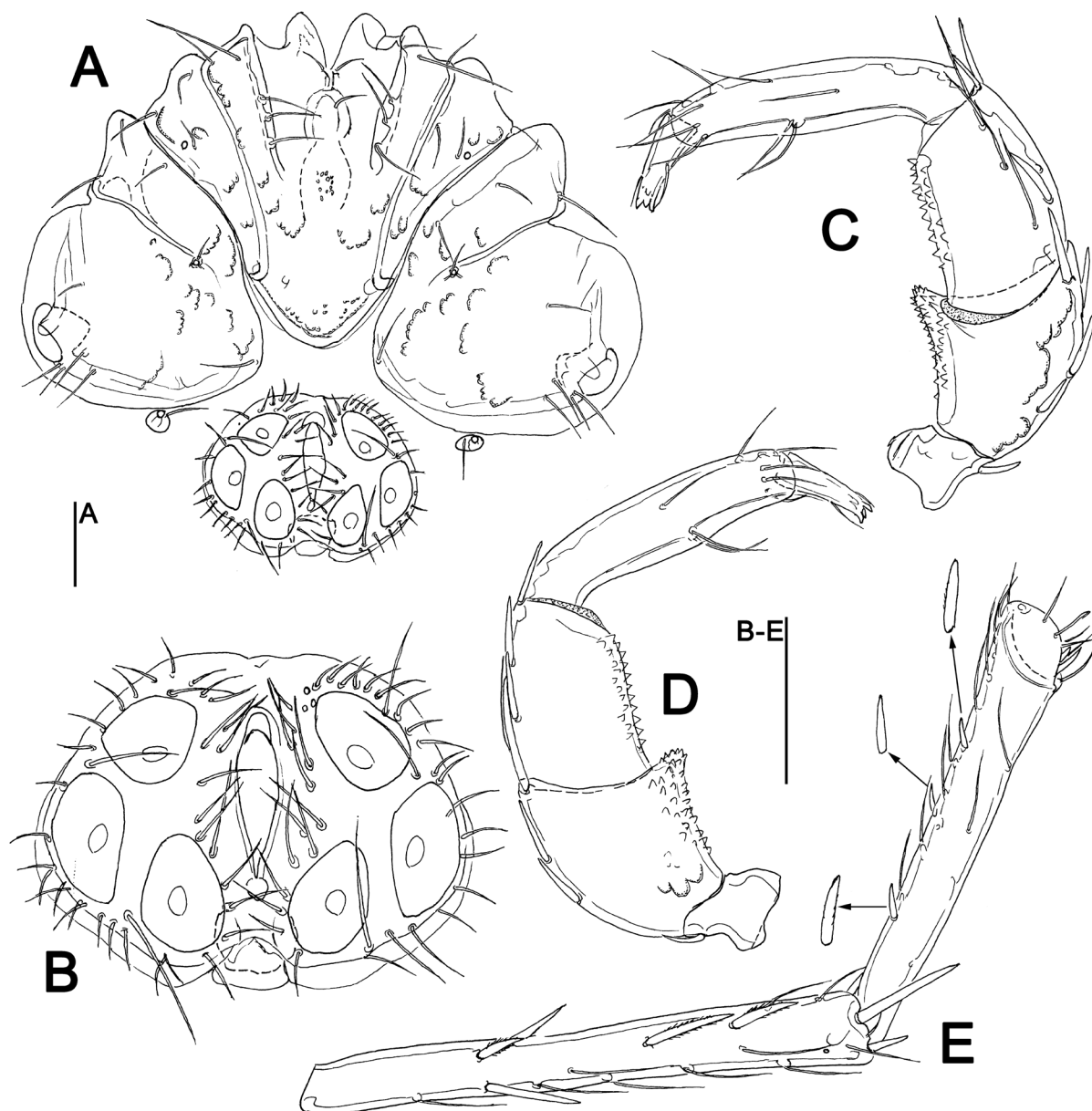


Figure 2. *Hygrobatès dorflingeri* n. sp., ♂ holotype [CCDB_45908_B05], Mesa Potamos, Cyprus: A – coxal and genital field; B – genital field; C-D – palp; E – IV-L-5 and -6 (inset: ventral peg-like setae on I-L-6 enlarged at 2x). Scale bars = 100 µm.

Discussion — Based on *COI* sequence data, the new species from Cyprus appears to be most closely related to *Hygrobatès turani*, a species described from a single male collected in a stream along the Turkish Black Sea coast (Pešić *et al.* 2026a). Males of *H. turani* can easily be distinguished from the new species by a more projecting posteromedial margin of Cx-I, an anterior margin of the genital plate bearing a distinct knob-shaped medial projection, and comparatively larger genital acetabula, L Ac-2 and -3 > 100 µm (see Pešić *et al.* 2026a).

Distribution — Cyprus; known only from the type locality (see Figure 6A in Pešić *et al.* 2026c).

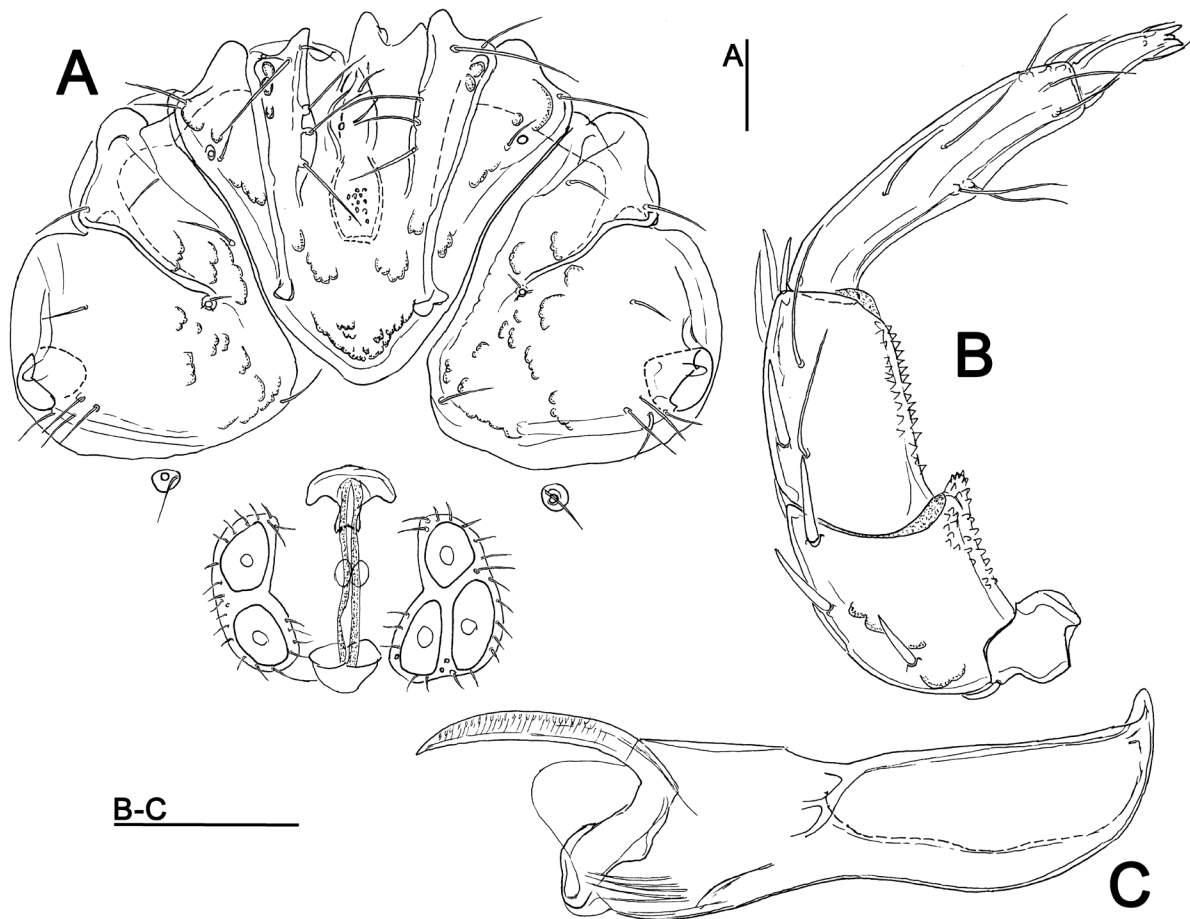


Figure 3. *Hygrobatos dorflingeri* n. sp., ♀ paratype [CCDB_45908_A08], Mesa Potamos, Cyprus: A – coxal and genital field; B – palp; C – chelicera. Scale bars = 100 µm.

***Hygrobatos ponticus cyprioticus* Pešić n. ssp.**

<https://www.zoobank.org/urn:lsid:zoobank.org:act:AE800B10-3347-4138-9889-E0C3CE7C6538>

Figures 4-5

Material examined — Holotype ♀, dissected and slide mounted (RMNH), sequenced (voucher Id: CCDB_45908_B03, sequence Id: CYPTU015-25), Cyprus, Limassol, Mesa Potamos stream, 34.876° N, 32.910° E, permanent stream, 23 May 2025, leg. Pešić, Tziortzis & Papatheodoulou. Paratypes: 1 ♂, same place and data as the holotype, dissected and slide mounted (RMNH); 1 ♀, sequenced (voucher Id: CCDB_45908_H02, sequence Id: CYPTU086-25), Paphos district, Stavros tis Psokas near Sarama Quarry, 34.952576° N, 32.520718° E, 21 May 2026, leg. Papatheodoulou; 2 ♀, Paphos district, Agriokalami Coast (river mouth), 34.81257° N, 32.39443° E, 20 May 2026, leg. Papatheodoulou, 1 ♀ sequenced (voucher Id: CCDB_45908_H06, sequence Id: CYPTU090-25).

Diagnosis — *Morphological*: P-2 with a triangular, bluntly pointed, relatively shortened (L/W ratio about 1.0) ventrodistal projection. Anterior margin of the genital field with a shallow indentation and a distinct knob-shaped medial projection in the centre of the indentation; Ac-2 and Ac-3 placed next to each other; female gonopore as long as the genital plates. *Molecular*: This lineage is represented by a unique Barcode Index Number (BIN: **BOLD:AH18370**). It differs from its closest lineages, *H. ponticus ponticus* and *H. ponticus yildirimi*, by 8.64% and 10.25% K2P divergence, respectively.

Description. General features — Colour yellow to brown, integument soft, strongly lineated. Posteromedial margin of Cx-I+II obtuse-triangular, caudo-lateral apodemes of Cx-I+II moderately developed; Cx-IV subtriangular, with a distinct nose-like protruding medial margin (Figure 4A). Acetabula in a triangular arrangement. P-2 ventral margin distally forming a triangular, bluntly pointed projection covered by small, scattered denticles; P-3 with denticles covering distal two-thirds of ventral

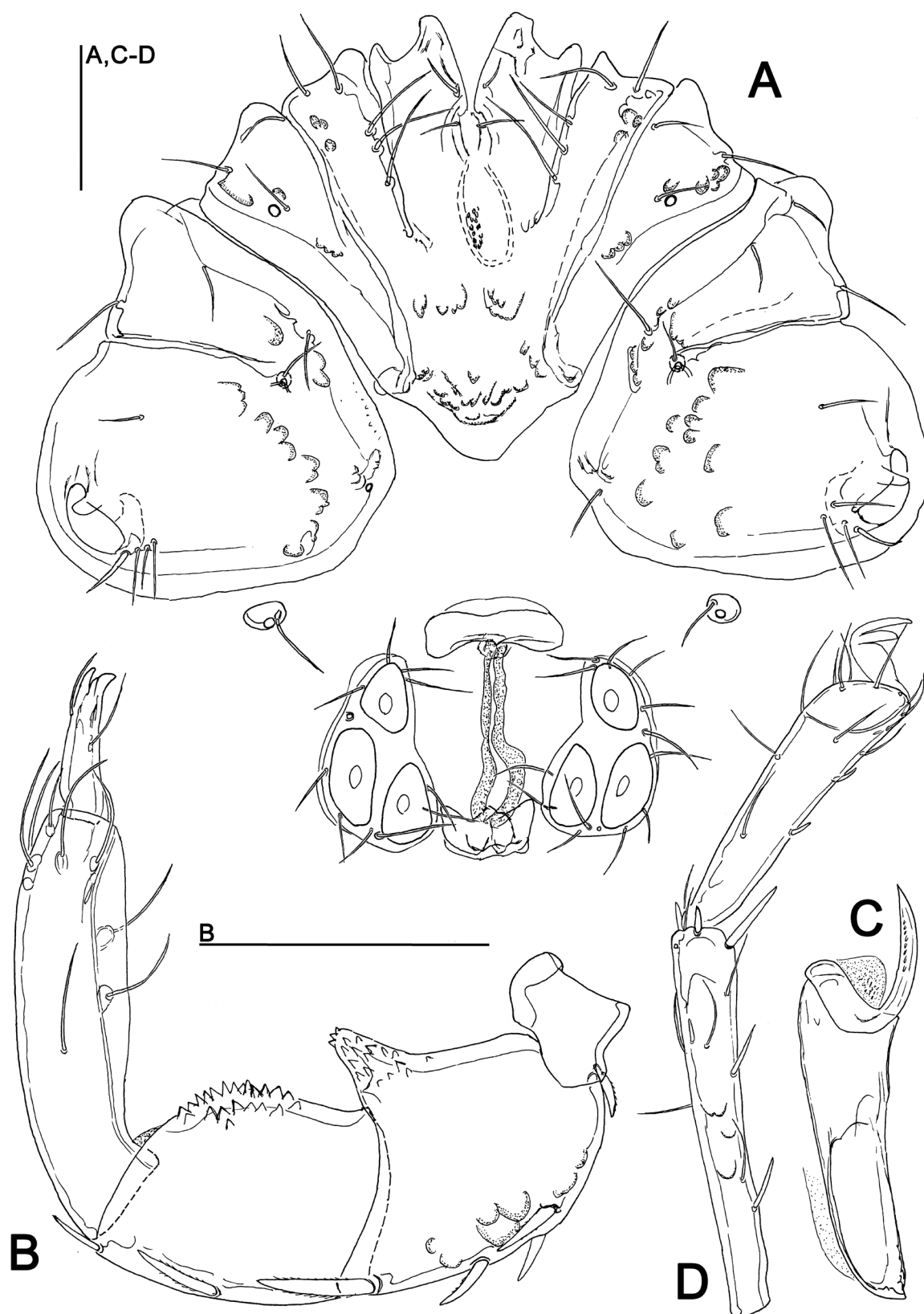


Figure 4. *Hygrobat es ponticus cypr ioticus* n. ssp., ♀ holotype [CCDB_45908_B03], Mesa Potamos, Cyprus: A – coxal and genital field; B – palp; C – chelicera; D – IV-L-5 and -6. Scale bars = 100 μm

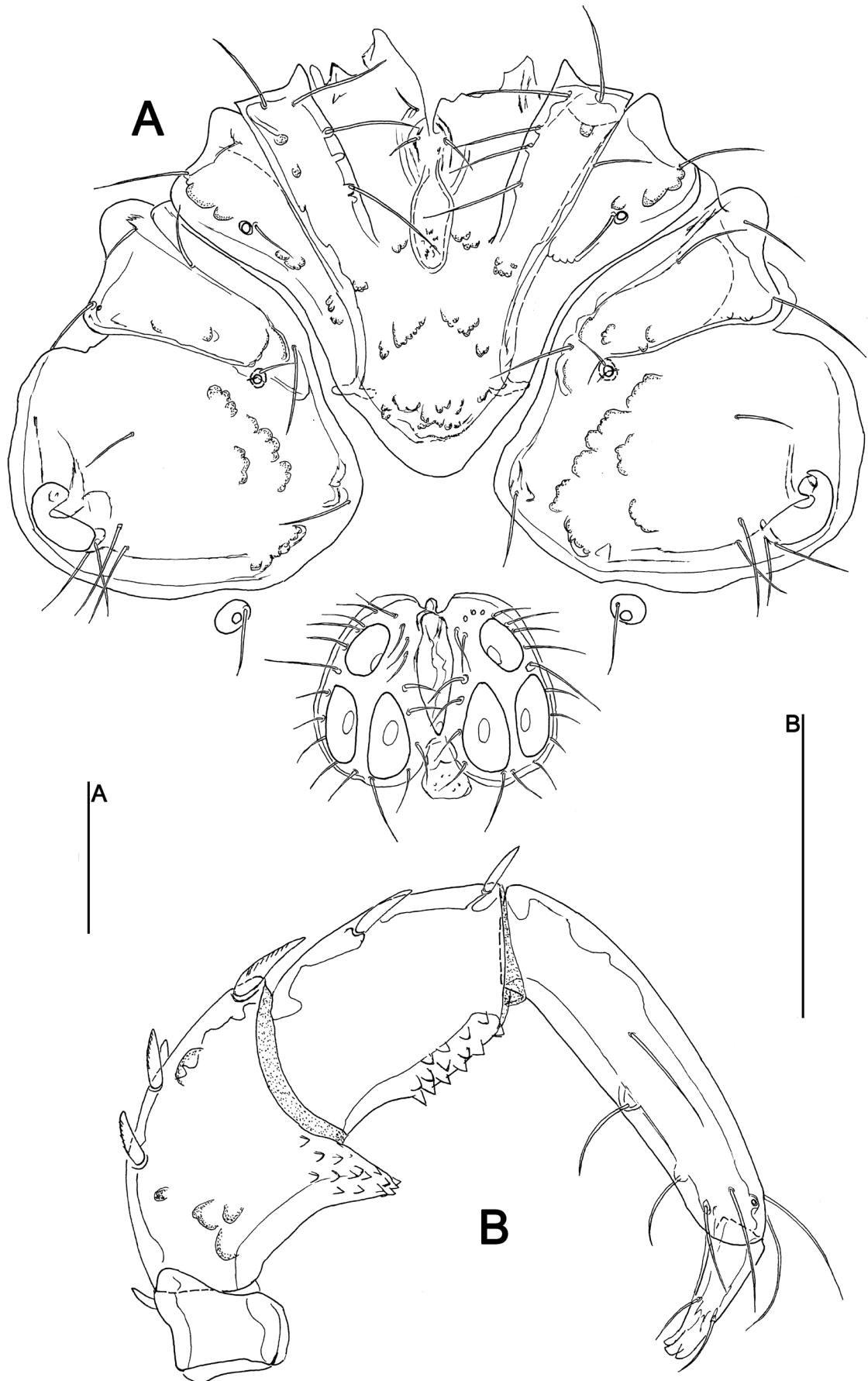


Figure 5. *Hygrobat es ponticus cyprioticus* n. ssp., ♂ paratype, Mesa Potamos, Cyprus: A – coxal and genital field; B – palp. Scale bars = 100 µm.

margin; P-4 ventral setae separated (Figures 4B, 5B). *Male* — Anterior margin of genital field with a shallow indentation and a small knob-shaped medial projection in the centre of indentation, genital plate typically with an irregular margin of secondary sclerotization, posterior margin of genital field deeply indented with a protrusion in the centre of the indentation, extending beyond the posterior margin, Ac-2 and -3 placed next to each other, Ac-3 subtriangular and elongated (Figure 4A). *Female* — Genital plates with a narrow border of porose secondary sclerotization, gonopore as long as genital plates (Figure 5A).

Measurements. Female (holotype [CCDB_45908_B03]) — Idiosoma L 850, W 678; coxal field: L 375; Cx-III 516; mL of Cx-I + gnathosoma L 312; distance between lateralmost ends of caudo-lateral Cx-II apodemes, 144; genital field L/W 181/236; genital plates L 126-134; pregenital sclerite W 95; gonopore L 150; L Ac 1-3: 48, 64-66, 56. Egg maximum diameter ($n = 2$) 147-150.

Palp — Total L 440; dL/H: P-1, 28/41; P-2, 117/77; P-3, 94/73; P-4, 150/39; P-5, 51/16; dL P-2/P-4 ratio 0.78. Chelicera total L 288, basal segment L 196, claw L 106, L claw/basal segment ratio 0.54.

Legs — dL of I-L-1-5: 69, 78, 97, 150, 166. dL of IV-L-1-6: 127, 116, 163, 246, 281, 225.

Male (paratype from Mesa Potamos stream; $n = 1$) — Idiosoma L 725; coxal field: L 355; Cx-III 463; mL of Cx-I + gnathosoma L 298; distance between lateralmost ends of caudo-lateral Cx-II apodemes, 146. Genital field L/W 135/161, ratio 0.84, L Ac-1-3: 39-41, 56, 56-60. Ejaculatory complex L 144.

Palp — total L 415; dL/H: P-1, 28/38; P-2, 109/70; P-3, 87/61; P-4, 143/34; P-5, 48/17; dL P-2/P-4 ratio 0.76.

Etymology — The subspecific epithet *cyprioticus* refers to Cyprus, where the subspecies was discovered. The name is an adjective in the nominative singular.

DNA Barcode Analysis and Species Delimitation — The *COI* sequences form a unique cluster (BIN: [BOLD:AH18370](#)), with the nearest neighboring BIN being that of *H. ponticus ponticus* ([BOLD:AE00649](#)). The genetic distance between these two lineages was estimated at 8.64% K2P divergence, while the distance to the other subspecies, *H. ponticus yildirimi*, was estimated at 10.25% K2P (Table 2).

Discussion — Molecular analysis revealed that the examined specimens from Cyprus represent a unique and well-supported *COI* clade. This clade appears to be genetically closer to the nominotypical subspecies from eastern Türkiye and Georgia than to *H. ponticus yildirimi*, which was originally described from southwestern Türkiye (Pešić *et al.* 2023). Given the allopatric distribution of the Cypriot taxon relative to the two mainland subspecies, alongside its substantial genetic divergence (8.6 to 10.3% K2P), we consider its status as a distinct subspecies justified.

Morphologically, the new subspecies can be separated from both mainland subspecies. Females of the new subspecies exhibit a comparatively shorter gonopore, whereas it is distinctly longer and exceeds the length of the genital plate in the other two subspecies (see Pešić *et al.* 2022, fig. 10A; Pešić *et al.* 2023, fig. 1A). In males, the posterior margin of the genital plate has a small central protrusion within the indentation that extends beyond the posterior margin of the genital field, whereas it does not exceed the posterior margin in the other two subspecies (see Pešić *et al.* 2022, fig. 10C; Pešić *et al.* 2023, fig. 1B). Furthermore, according to its original description, *Hygrobates ponticus yildirimi* tends to have a longer ventrodistal projection on P-2 than the other two subspecies (L/W ratio: 1.2 vs. 1.0 in *H. p. ponticus* and *H. p. cyprioticus* n. ssp.).

Distribution — Cyprus.

Biogeographical Remarks

Although the number of known water mite species from Cyprus is relatively limited (see Pešić *et al.* 2026c for an overview), several significant characteristics of the water mite community and its faunal similarities with neighboring regions can be highlighted. The proximity of the island to mainland Anatolia appears to have facilitated colonization by species originating from the latter region.

- *Torrenticola baueri* — Insular populations of this species belong to the same BIN that is widely distributed across mainland Anatolia and Iran (Pešić *et al.* 2025). Based on current knowledge, the Cypriot populations represent the westernmost limit of the species' distribution.
- *Atractides nodipalpis* complex — The *A. nodipalpis*-like populations from Cyprus belong to *A. concavus* Pešić & Smit, 2021, a species widely distributed across mainland Anatolia. This case represents a clear example of allopatric divergence at population level driven by geographic isolation. The Cyprus populations form a distinct Barcode Index Number (BIN: [BOLD:AH18403](#)) with a 4.27% *p*-distance from the closest mainland BIN ([BOLD:AEK3669](#)), which includes individuals of *A. concavus* from southwestern Anatolia. While long-term insular isolation has resulted in a clear genetic break, evolutionary pressures likely have not yet caused these populations to diverge into a distinct species.
- *Hygrobatas persicus* — Recently, Pešić *et al.* (2026b) argued that colonization by *H. persicus* occurred significantly earlier than the final separation of Cyprus from the mainland (south-central Anatolia), which took place during the Pleistocene–Holocene transition within the last 2.6 million years (Yılmaz 2021). These authors suggested that the *H. persicus* populations on the island have likely not had sufficient evolutionary time to diverge into a distinct species.
- *Hygrobatas calliger* complex — High genetic distances indicate that both new taxa described in this study are the result of long-term genetic isolation. Both taxa have their closest congeneric relatives in mainland Türkiye (Anatolia), further supporting the hypothesis that ancestral lineages colonized the island well before its final separation from the mainland.

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